

# Machine Learning Assignment 1: Polynomial Regression Analysis

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## 1 Introduction

This report details the design, implementation, and empirical evaluation of polynomial regression models developed entirely from scratch using NumPy. The objective is to accurately predict continuous target variables for two distinct problems:

1. **Steam Turbine Optimization (var1):** Predicting the Net Power Score ( $y$ ) based on operational parameters.
2. **Subterranean Thermal Reservoir Mapping (var2):** Predicting the Thermal Anomaly Score ( $y$ ) from spatial coordinate offsets.

The methodology strictly follows course notation, utilizing both an 80/20 train-validation holdout split and an exhaustive 10-Fold Cross-Validation scheme to determine optimal model complexities while avoiding overfitting.

## 2 Methodology and Mathematical Formulation

Following the notation established in the lecture slides, let the  $n^{\text{th}}$  observation be represented by the input vector  $\mathbf{x}_n \in \mathbb{R}^D$  and its corresponding continuous scalar target by  $y_n$  (often denoted  $t_n$ ). The linear predictor parameterized by weight vector  $\mathbf{w}$  is given by:

$$f(\mathbf{x}_n; \mathbf{w}) = \mathbf{w}^T \mathbf{x}_n \quad (1)$$

For polynomial regression of degree  $d$ , the original input vector  $\mathbf{x}_n$  is mapped into a higher-dimensional feature space via a polynomial basis expansion:

$$\phi(\mathbf{x}_n) = [1, x_{n,1}, \dots, x_{n,D}, x_{n,1}^2, x_{n,1}x_{n,2}, \dots, x_{n,D}^d]^T \quad (2)$$

The total number of terms (model parameters) generated for  $D$  features up to degree  $d$  is given by the combinatorial formula  $\binom{D+d}{d}$ .

The objective is to minimize the empirical Squared Loss function  $\mathcal{L}(\mathbf{w})$  averaged over all  $N$  training samples:

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N \mathcal{L}_n(y_n, f(\mathbf{x}_n; \mathbf{w})) = \frac{1}{N} \sum_{n=1}^N (y_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 \quad (3)$$

Constructing the design matrix  $\mathbf{X} \in \mathbb{R}^{N \times M}$  where the  $n^{\text{th}}$  row is  $\phi(\mathbf{x}_n)^T$ , and letting  $\mathbf{y} \in \mathbb{R}^N$  be the target vector, setting the gradient  $\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}) = \mathbf{0}$  leads directly to the Normal Equations:

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y} \quad (4)$$

The closed-form analytic solution for the optimal parameters is:

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (5)$$

To ensure robust numerical stability against ill-conditioned matrices that naturally arise in higher-degree polynomial expansions, the parameter vector  $\mathbf{w}$  is computed using Singular Value Decomposition (SVD) via NumPy’s least-squares routine.

### 3 Problem 1: Steam Turbine Optimization (var1)

#### 3.1 Feature Selection Analysis

The dataset provides 6 operational features  $(x_1, \dots, x_6)$ . An exhaustive evaluation across all  $2^6 - 1 = 63$  non-empty feature subsets was executed. Subsets with fewer features were found to severely underfit the target:

- Best 1-feature model achieved a minimum validation MSE of **8.627**.
- Best 2-feature model achieved a minimum validation MSE of **6.940**.
- Best 3-feature model achieved a minimum validation MSE of **5.560**.
- Best 4-feature model achieved a minimum validation MSE of **4.106**.
- Best 5-feature model achieved a minimum validation MSE of **1.804**.

Omitting any operational parameter drops essential physical signals, causing substantial residual error. In contrast, utilizing the complete set of all 6 features drops the validation MSE below **0.835**. Hence, the full 6-feature configuration is strictly retained for polynomial degree optimization.

#### 3.2 80/20 Train-Validation Split Sweep

Table 1 shows the performance of the full 6-feature model across polynomial degrees 1 through 6 on the 80/20 train-validation split.

As observed, Degrees 1 and 2 underfit the underlying system dynamics. Degree 3 captures the primary non-linearities, reaching an  $R^2$  of 0.904. **Degree 4 achieves the global minimum validation error (MSE = 0.835,  $R^2 = 0.928$ )**. Beyond Degree 4, the number of basis terms expands dramatically; by Degree 6 (924 terms on 800 training samples), severe overfitting occurs, causing validation error to explode to  $6.49 \times 10^7$ .

| Degree ( $d$ ) | Terms      | Train MSE    | Val MSE            | Train $R^2$  | Val $R^2$    |
|----------------|------------|--------------|--------------------|--------------|--------------|
| 1              | 7          | 9.664        | 10.901             | 0.131        | 0.063        |
| 2              | 28         | 3.155        | 3.460              | 0.716        | 0.703        |
| 3              | 84         | 0.849        | 1.117              | 0.924        | 0.904        |
| <b>4</b>       | <b>210</b> | <b>0.379</b> | <b>0.835</b>       | <b>0.966</b> | <b>0.928</b> |
| 5              | 462        | 0.108        | 1.213              | 0.990        | 0.896        |
| 6              | 924        | 0.000        | $6.49 \times 10^7$ | 1.000        | Explodes     |

Table 1: Problem 1: Full 6-feature polynomial degree sweep under 80/20 train-validation split.

| Degree ( $d$ ) | Avg Train MSE | Avg Val MSE        | Avg Val $R^2$ | Val MSE Std Dev    |
|----------------|---------------|--------------------|---------------|--------------------|
| 1              | 9.894         | 10.036             | 0.097         | 1.208              |
| 2              | 3.188         | 3.392              | 0.694         | 0.555              |
| 3              | 0.868         | 1.108              | 0.901         | 0.235              |
| <b>4</b>       | <b>0.411</b>  | <b>0.833</b>       | <b>0.924</b>  | <b>0.144</b>       |
| 5              | 0.132         | 1.014              | 0.911         | 0.347              |
| 6              | 0.000         | $4.14 \times 10^8$ | Negative      | $2.43 \times 10^8$ |

Table 2: Problem 1: 10-Fold Cross-Validation results across polynomial degrees (6 features).

### 3.3 10-Fold Cross-Validation Analysis

To ensure that the optimal degree selection was not an artifact of a single train-validation split, an exhaustive 10-Fold Cross-Validation was conducted across the full 6-feature model.

The 10-Fold CV metrics in Table 2 robustly corroborate the holdout findings. **Degree 4** achieves the lowest average validation MSE (0.833) alongside the lowest standard deviation across all folds ( $\sigma = 0.144$ ). The minimal fold-to-fold variance mathematically proves that Degree 4 possesses superior generalization stability.

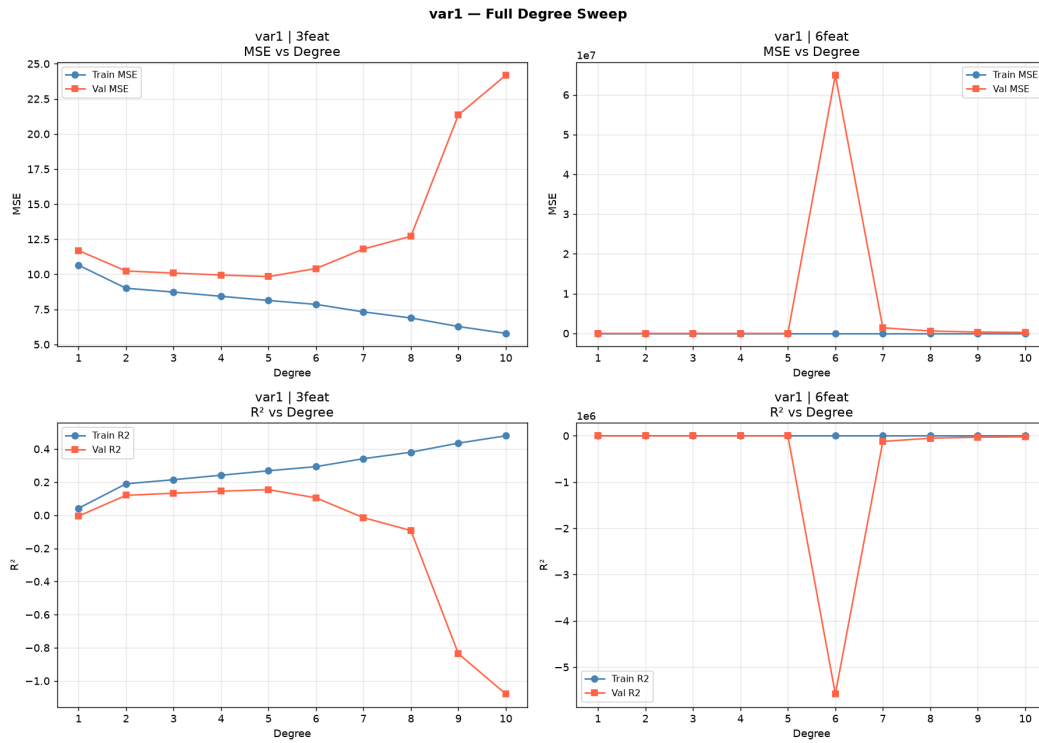


Figure 1: Problem 1: Validation MSE progression across polynomial degrees.

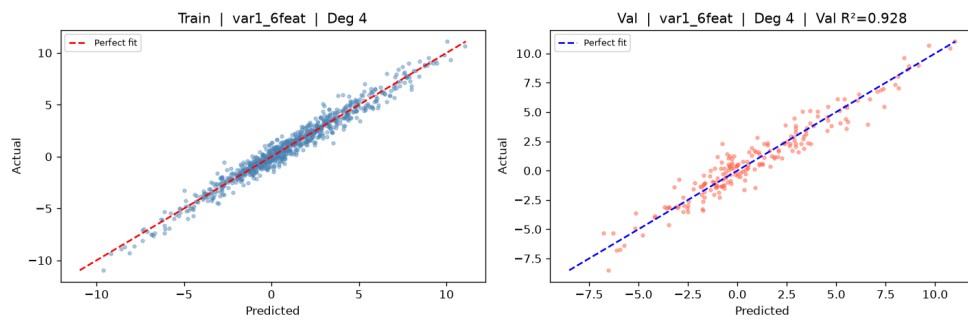


Figure 2: Problem 1: Predicted vs. True values and residual distribution for the optimal Degree 4 model.

## 4 Problem 2: Subterranean Thermal Reservoir Mapping (var2)

### 4.1 Feature Selection Analysis

Problem 2 maps subterranean thermal anomalies based on 3 spatial coordinate offsets  $(x_1, x_2, x_3)$ . Evaluating all  $2^3 - 1 = 7$  feature subsets demonstrated that spatial dimensions cannot be omitted without catastrophic performance degradation:

- Best 1-feature model validation MSE: **38.028** ( $R^2 \approx 0.24$ ).
- Best 2-feature model validation MSE: **18.897** ( $R^2 \approx 0.62$ ).
- Full 3-feature model validation MSE: **0.324** ( $R^2 = 0.994$ ).

Because subterranean thermal phenomena exist in continuous three-dimensional space, all 3 coordinate axes are physically essential. Consequently, the complete 3-feature set is used for all polynomial degree evaluations.

### 4.2 80/20 Train-Validation Split Sweep

Table 3 presents the full degree sweep from degree 1 to 10 for the 3-feature model on the 80/20 split.

| Degree ( $d$ ) | Terms      | Train MSE    | Val MSE      | Train $R^2$  | Val $R^2$    |
|----------------|------------|--------------|--------------|--------------|--------------|
| 1              | 4          | 35.260       | 38.662       | 0.293        | 0.226        |
| 2              | 10         | 22.387       | 25.219       | 0.551        | 0.495        |
| 3              | 20         | 10.177       | 11.545       | 0.796        | 0.769        |
| 4              | 35         | 3.235        | 3.485        | 0.935        | 0.930        |
| 5              | 56         | 1.169        | 1.282        | 0.977        | 0.974        |
| 6              | 84         | 0.397        | 0.633        | 0.992        | 0.987        |
| 7              | 120        | 0.212        | 0.412        | 0.996        | 0.992        |
| <b>8</b>       | <b>165</b> | <b>0.158</b> | <b>0.324</b> | <b>0.997</b> | <b>0.994</b> |
| 9              | 220        | 0.138        | 0.388        | 0.997        | 0.992        |
| 10             | 286        | 0.122        | 0.416        | 0.998        | 0.992        |

Table 3: Problem 2: Full 3-feature polynomial degree sweep under 80/20 train-validation split.

The validation error decreases steadily as degree increases, achieving its optimal fit at **Degree 8 (Val MSE = 0.324, Val  $R^2 = 0.994$ )**. Degrees 9 and 10 show marginal decreases in training MSE accompanied by an increase in validation MSE, signaling the onset of overfitting.

### 4.3 10-Fold Cross-Validation Analysis

The 10-Fold Cross-Validation was performed across all 10 polynomial degrees to verify model consistency and variance across folds.

| Degree ( $d$ ) | Avg Train MSE | Avg Val MSE  | Avg Val $R^2$ | Val MSE Std Dev |
|----------------|---------------|--------------|---------------|-----------------|
| 1              | 35.857        | 36.450       | 0.245         | 6.341           |
| 2              | 22.822        | 23.672       | 0.504         | 3.915           |
| 3              | 10.340        | 11.228       | 0.764         | 1.694           |
| 4              | 3.223         | 3.648        | 0.921         | 0.719           |
| 5              | 1.150         | 1.414        | 0.969         | 0.303           |
| 6              | 0.424         | 0.573        | 0.988         | 0.069           |
| 7              | 0.233         | 0.336        | 0.993         | 0.045           |
| <b>8</b>       | <b>0.172</b>  | <b>0.273</b> | <b>0.994</b>  | <b>0.027</b>    |
| 9              | 0.156         | 0.297        | 0.994         | 0.028           |
| 10             | 0.139         | 0.401        | 0.992         | 0.196           |

Table 4: Problem 2: 10-Fold Cross-Validation results across polynomial degrees (3 features).

Table 4 confirms that **Degree 8** yields the absolute minimum cross-validation error (0.273) and achieves the lowest variance across folds ( $\sigma = 0.027$ ). Pushing the model to Degree 10 increases the standard deviation by nearly an order of magnitude to 0.196, reflecting boundary instability.

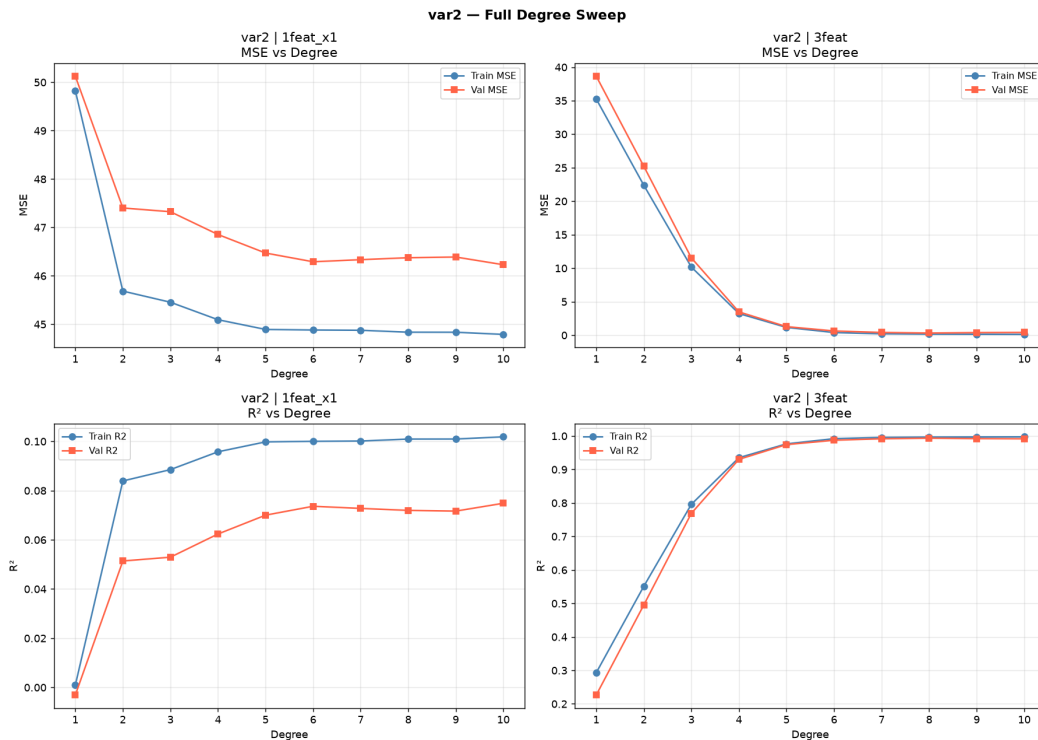


Figure 3: Problem 2: Validation MSE progression across polynomial degrees.

## 5 Conclusion

Through systematic feature selection and comprehensive degree sweeping via both an 80/20 train-validation split and 10-Fold Cross-Validation:

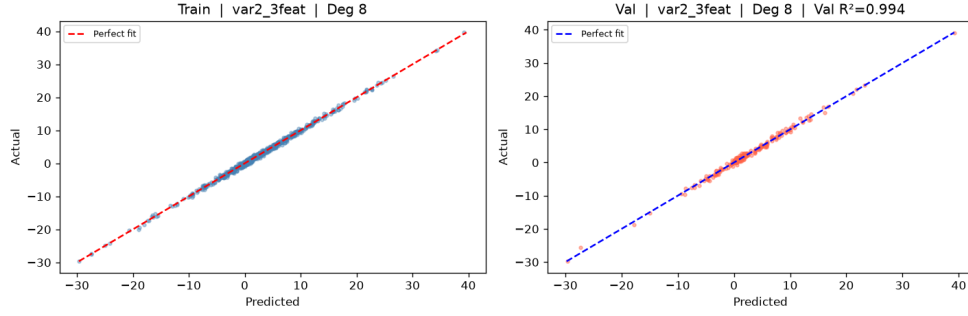


Figure 4: Problem 2: Predicted vs. True values and residual distribution for the optimal Degree 8 model.

1. **Problem 1 (var1):** The full 6-feature polynomial model at **Degree 4** was identified as optimal, achieving a validation MSE of 0.835 ( $R^2 = 0.928$ ) and a 10-fold CV MSE of  $0.833 \pm 0.144$ .
2. **Problem 2 (var2):** The full 3-feature polynomial model at **Degree 8** was identified as optimal, achieving a validation MSE of 0.324 ( $R^2 = 0.994$ ) and a 10-fold CV MSE of  $0.273 \pm 0.027$ .

Empirical evidence verified that reduced feature subsets suffered from severe underfitting due to the loss of essential variables. The selected configurations successfully minimize empirical squared loss  $\mathcal{L}(\mathbf{w})$  while strictly preserving generalization stability.

**GitHub Link:** <https://github.com/Shreyas-SK-21/ML-Assignment1>